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Serendipity

in which astronomers are forced to conclude,
without any prior predictions,
that black holes a millionfold heavier than the Sun
inhabit the cores of galaxies (probably)

Radio Galaxies

If, in 1962 (when theoretical physicists were just beginning to accept the concept of a black hole), anyone had asserted that the Universe contains gigantic black holes, millions or billions of times heavier than the Sun, astronomers would have laughed. Nevertheless, astronomers unknowingly had been observing such gigantic holes since 1939, using radio waves. Or so we strongly suspect today.

Radio waves are the opposite extreme to X-rays. X-rays are electromagnetic waves with extremely short wavelengths, typically 10,000 times *shorter* than the wavelength of light (Figure P.2 in the Prologue). Radio waves are also electromagnetic, but they have long wavelengths, typically a few meters from wave crest to wave crest, which is a million times *longer* than the wavelength of light. X-rays and radio waves are also opposites in terms of wave/particle duality (Box 4.1)—the propensity of electromagnetic waves to behave sometimes like a wave and sometimes like a particle (a photon). X-rays typically behave like high-energy particles (photons) and thus are most easily detected with Geiger counters in which the X-ray photons hit atoms, knocking electrons off them (Chapter 8). Radio waves almost always behave like waves of electric and magnetic force, and thus are most easily detected with wire or metal antennas in